VHDL Implementation of NIST Ascon AEAD128 on a Nexys A7 FPGA

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Abstract

This paper describes a VHDL-2008 implementation of the standardized NIST Ascon-AEAD128 algorithm for a Nexys A7 100T FPGA. The design uses one shared 320-bit permutation datapath with configurable round unrolling and optional input/output queues. A 128-bit streaming interface accepts associated data and payload bytes, including empty and partial blocks. Encryption returns ciphertext and a tag. Decryption returns provisional plaintext that an external quarantine releases only after tag verification. The portable test flow passed all 1,089 official known-answer vectors in both directions with GHDL and Vivado XSim. Generated Verilog was also checked with Icarus Verilog and Verilator. Vivado 2023.2 completed synthesis, placement, routing, timing analysis, and bitstream generation. At a simulated 100 MHz, the complete 2,397,696-byte FP32 adapter payload encrypts and decrypts in 449,581 cycles per direction with folded absorption, compared with 899,153 for the buffered three-round core and 1,798,305 for the one-round baseline. All recovered bytes match and authentication passes. The folded core routes at 100 MHz with unconstrained external I/O; a board-integrated copy also meets 100 MHz in one run. Earlier physical captures validate only the buffered design. Later multicore board tests pass a 3,072-byte fixture for one, two and four folded cores; full-file physical transfer remains unmeasured. A separate dependent-round pipeline experiment finds that controller fixes improve the unchanged 12-clock design from 70.8 to 49.8 ns per block, while two- and three-stage round splits increase elapsed latency. Sections XIV through XVI distinguish simulation, routed timing, modeled memory and physical fixture evidence.

I. Research objective

The goal of this work is to build an understandable FPGA implementation of authenticated encryption. The core must follow the final standard, handle normal streaming stalls, and make authentication failure visible to the surrounding system. The first version favors a clear control path over maximum throughput. It is meant to support later research on faster datapaths and hardware security countermeasures without changing the external transaction format.

II. Ascon-AEAD128 background

NIST SP 800-232 defines Ascon-AEAD128 with a 128-bit key, 128-bit nonce, 128-bit authentication tag, 128-bit data rate, and a 320-bit internal state [1]. The state is divided into five 64-bit words. Initialization and finalization use a 12-round permutation. Associated-data and full payload blocks use an 8-round permutation. Each round applies a constant to one state word, evaluates 64 five-input S-box slices in parallel, and then performs word-specific rotations and XOR operations.

The final NIST version was selected instead of legacy Ascon-128 or Ascon-128a. The final standard changed the initialization value and changed byte mapping to little endian [1]. For this core, byte lane 0 is bits 7 through 0 of the 128-bit port. Padding places the byte value 01 immediately after the last valid byte. Using the official mapping avoids a common result where a design is internally consistent but does not match current NIST vectors.

Initialize	Absorb AD	Domain	Process data	Finalize	**Tag**
p12	p8	separate	p8	p12	**128 bits**

Fig. 1. Ascon-AEAD128 processing phases, adapted from NIST SP 800-232 [1].

III. Original hardware architecture

The controller owns a 320-bit transaction state S, organized as five 64-bit words. It handles byte masks, padding, key additions, domain separation and tag comparison. A separate permutation engine owns a second 320-bit register P, a combinational round and a round counter (Fig. 2). A start request loads S into P. After eight or twelve round clocks, the engine raises a registered done signal and the controller copies P back into S. The same engine serves initialization, AD, payload and finalization. Each call therefore includes a load clock and a completion-copy clock in addition to its round clocks. Section XIV removes this handoff from the unrolled design.

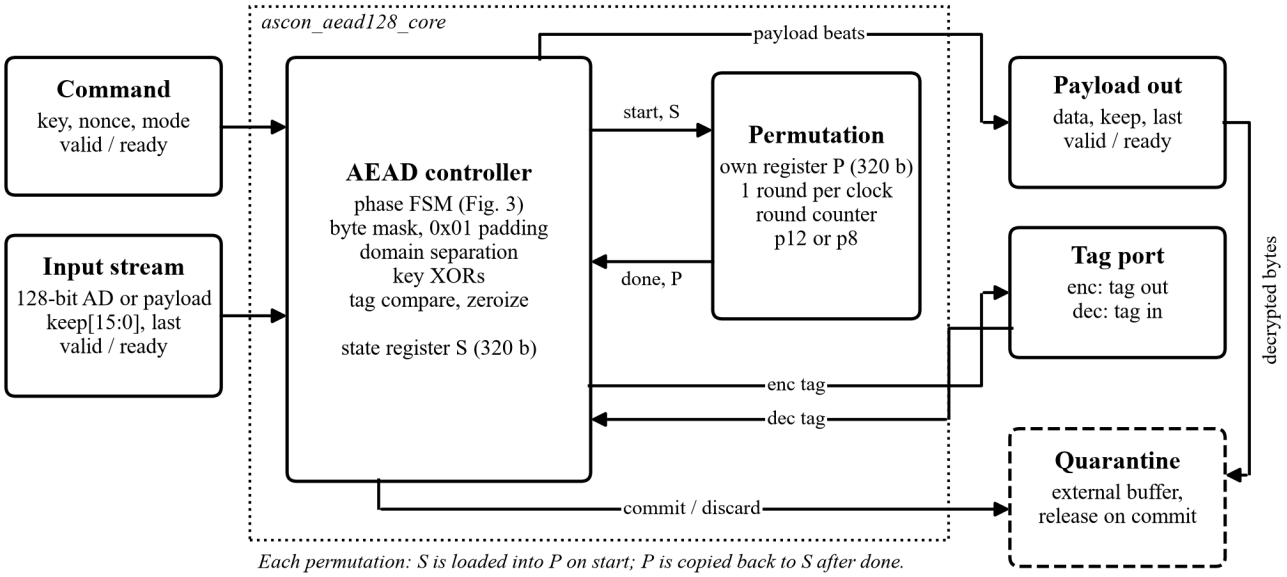


Fig. 2. Original one-round architecture. The controller owns S and the permutation engine owns P, both 320 bits. Their registered start/done interface transfers the state between them. Decrypted bytes remain provisional until an external quarantine receives commit.

The controller accepts a command containing the key, nonce, and direction. Associated data must finish before payload begins. Each stream beat includes 16 keep bits and a last flag. Keep bits must start at lane 0 and remain contiguous. An empty phase uses a terminal beat with no keep bits set. Any invalid ordering or mask produces a protocol error and sends all transaction storage through the same zeroization state used after a normal result.

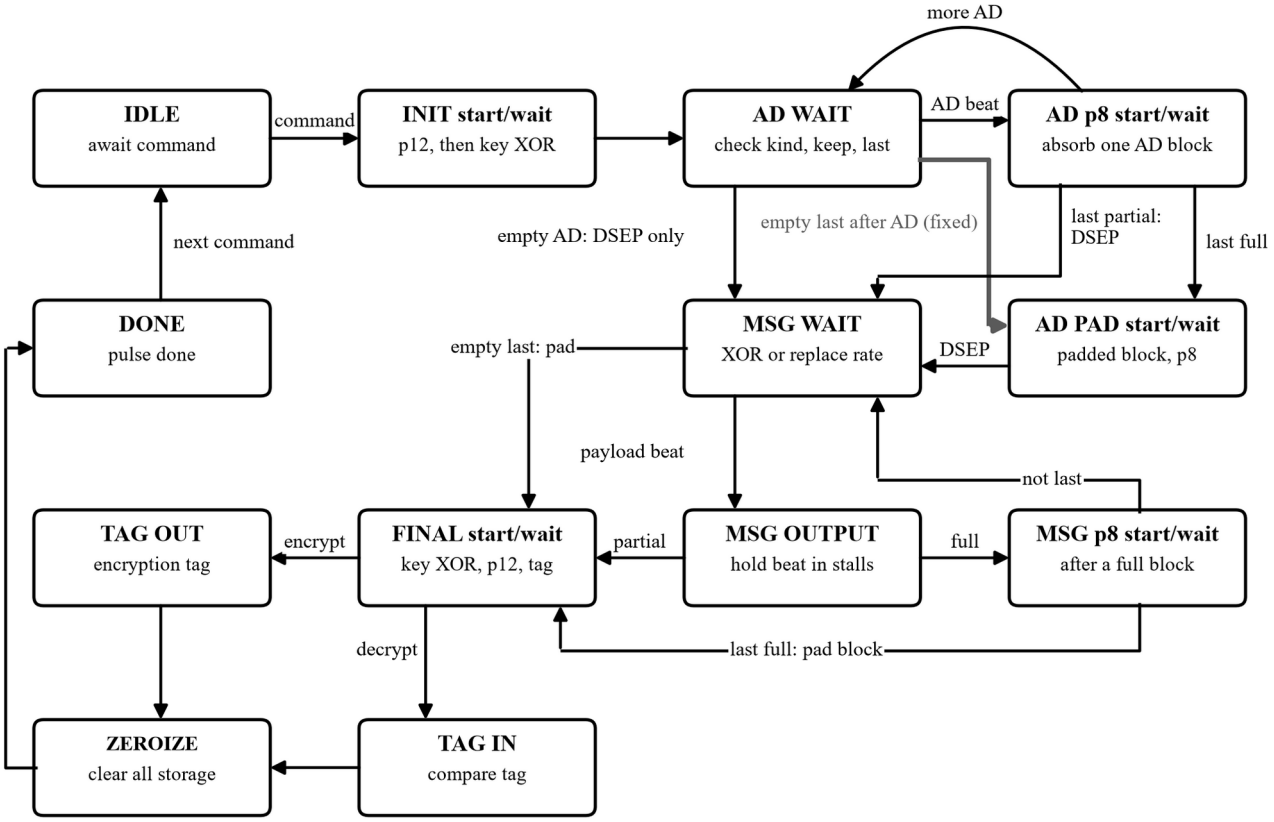
IV. Authentication-safe decryption

A decryption core cannot know whether plaintext is authentic until it processes the final tag. Storing every possible message inside the cryptographic core would make its memory size depend on the application. This design sends tentative plaintext to an external quarantine buffer instead. A correct tag pulses commit. A wrong tag pulses discard and never pulses commit. The interface therefore makes the safe release rule explicit, but the system integrator must enforce it.

V. Original controller and verification

The finite-state machine was made for this streaming architecture. Start states issue a one-cycle request to the permutation engine. Wait states prevent another request until the current permutation ends. Output states hold data stable during backpressure. Tag states either wait for the encryption tag to be accepted or wait for a decryption tag to be supplied.

The AD path must distinguish empty AD from an empty beat that terminates earlier full AD blocks. Non-empty AD is padded with 0x01 and zeros to a multiple of the 16-byte rate. An exact boundary therefore requires a separate padding-only block and its p8; genuinely empty AD needs only domain separation [1]. Fig. 3 includes the corrected empty-terminator transition. A full final payload block also needs an empty padded block, but that block is combined with the finalization key before p12 rather than followed by another p8.



START states request one permutation; WAIT states hold until its registered done. TAG IN pulses commit or discard. Protocol errors in AD WAIT or MSG WAIT go to ZEROIZE (not drawn). Gray: transition added by the AD padding fix.

Fig. 3. Original controller after the AD-padding fix. START and WAIT pairs are grouped for readability. The gray edge handles an empty terminator after non-empty AD. Protocol errors in AD WAIT or MSG WAIT enter ZEROIZE; TAG IN compares the tag and pulses commit or discard.

Verification began with an independent Python model. The model was compared with the official ascon-c known-answer file pinned to a recorded source revision [2]. Cocotb then drove the same vectors through the VHDL core. The full run covered all 1,089 combinations of associated-data and plaintext lengths from 0 through 32 bytes, with encryption followed by decryption. Separate tests stalled each output, reset an active transaction, sent invalid masks and phase ordering, checked fixed-seed messages up to 80 bytes, and changed the key, nonce, associated data, ciphertext, and tag.

Verification level	Cases	Result	Evidence
Python reference	81 boundary pairs	PASS	official KAT comparison
GHDL direct VHDL	1,089 × 2	PASS	encrypt and decrypt
Protocol/security	9 test groups	PASS	stalls, reset, zeroization
GHDL TextIO	1,089 × 2	PASS	two output text files
Vivado XSim	1,089 × 2	PASS	official KAT TextIO bench
Icarus netlist	1,089 × 2	PASS	GHDL-generated Verilog
Verilator netlist	1,089 × 2	PASS	GHDL-generated Verilog
Vivado implementation	placed and routed	PASS	reports and bitstream
Nexys A7 board	four demo vectors	PENDING	programming and hardware test

Table I. Verification and implementation status as of September 18, 2026.

VI. Vivado flow and FPGA demonstration

The repository contains Tcl scripts for Vivado 2023.2 and an XDC file for the Nexys A7 100T part xc7a100tcsg324-1. The constraint sets the board clock to 100 MHz. A pure VHDL XSim testbench writes simulation_results.txt for human review and simulation_vectors.txt for scripts. The same file-driven bench was checked with GHDL across all 1,089 vectors in both directions. XSim used the identical TextIO input and completed 2,178 encryption and decryption checks with no failures. The official-vector TextIO run measured 34 to 88 command-to-done cycles. For a 32-byte payload, the mean across all 33 associated-data lengths was 72.636 cycles, equal to 352.441 Mb/s at 100 MHz. These are cycle-based simulation results, not a measured maximum clock rate. The implementation results were extracted from the routed Vivado reports. Vivado used 1717 slice LUTs and 1372 slice registers, with no block RAM. The routed design met the 100 MHz constraint with 2.941 ns worst setup slack. Vivado estimated 0.137 W total on-chip power and generated the FPGA bitstream. Physical board programming and demonstration testing were pending at this baseline stage; Section XIII records the later board tests. The original routed report included a CFGBVS-1 configuration warning. The repository XDC now sets CFGBVS to VCCO and CONFIG_VOLTAGE to 3.3 for the Nexys A7 configuration bank. Physical-board validation was pending at this baseline stage; later results appear in Section XIII. The power value is a Vivado estimate rather than a board measurement.

The board wrapper stores four official vectors: empty input, a one-byte partial block, one full block, and a 17-byte two-block case. Switches choose the vector, encryption or decryption, payload or tag display, block number, and byte number. The center button starts a run and the upper button resets the design. LEDs display the selected result byte, busy and done state, authentication pass or fail, and a protocol error.

VII. WSL Ubuntu environment

An application experiment completed on October 6, 2026 used trained GPT-2 weight bytes as the input to the existing Ascon-AEAD128 VHDL core. The workflow placed GPT-2 training in Ubuntu under Windows Subsystem for Linux 2 (WSL 2) and retained Vivado 2023.2 on Windows for RTL simulation. This allowed the Linux training process to use the laptop NVIDIA GPU without a separate desktop virtual machine or a second Vivado installation [7], [8].

Component	Recorded configuration
Host	Windows 11 Home; GIGABYTE AORUS 15P YD
CPU and memory	Intel Core i7-11800H; 8 cores, 16 logical processors; 32 GB RAM
GPU	NVIDIA GeForce RTX 3080 Laptop GPU; 8,192 MiB VRAM
Linux environment	Ubuntu 22.04.5 LTS under WSL 2; user kroix
Python and PyTorch	Python 3.10; PyTorch 2.11.0+cu128
Other libraries	Transformers 5.18.0; NumPy 1.26.4
GPU driver	Windows NVIDIA driver 576.52; nvidia-smi reports CUDA capability 12.9
Hardware tool	Vivado 2023.2 and XSim on Windows

Table II. Software and hardware environment recorded for the GPT-2 experiment.

Ubuntu was installed from an administrator PowerShell session with wsl --install -d Ubuntu-22.04. The distribution was confirmed as WSL version 2, and nvidia-smi inside Ubuntu detected the RTX 3080. WSL uses the Windows NVIDIA driver; a separate Linux NVIDIA display driver was not installed [8]. The cu128 suffix identifies the PyTorch CUDA 12.8 build. The CUDA version printed by nvidia-smi describes driver capability and does not by itself establish an installed CUDA compiler.

The project was cloned into ~/projects/SAGE_Accelerator/llmc-gpt2, and a Python virtual environment named .venv kept its dependencies separate. Git, build-essential, python3-pip, and python3-venv supplied the basic development tools. GPU access was verified by allocating a 256 by 256 tensor on CUDA, multiplying it, and synchronizing the device. This calculation passed before training began. The native C/CUDA trainer was not used for GPU training; the Makefile reported that nvcc, MPI, and NCCL development support were unavailable. These components were unnecessary for the PyTorch run and CPU correctness test.

Appendix A records the shell commands and final export code. The setup and measurements below describe this specific small experiment; they do not establish production model quality or physical FPGA performance.

VIII. GPT-2 training and weight export

The software was taken from SPIRE-GMU/SAGE_Accelerator, llmc-gpt2, at commit 4e586d80b61ad7577c64391202a8c43104728f1d [9]. This folder contains an llm.c implementation and the PyTorch reference trainer train_gpt2.py. The pretrained gpt2 model was loaded through Hugging Face. The run fine-tuned existing GPT-2 weights for 40 optimizer updates; it did not train GPT-2 from random initialization.

The Tiny Shakespeare preparation script downloaded approximately 1.06 MiB of text and generated two token files. The training split contained 305,260 tokens in 611,544 bytes, and the validation split contained 32,768 tokens in 66,560 bytes. The binary sizes include a 1,024-byte dataset header. Training used the training file and validation used the separate validation file.

Before fine-tuning, a zero-update run with batch size 4 and sequence length 64 generated the pretrained FP32 and BF16 checkpoints, tokenizer, and PyTorch debug state. The CPU test was built with make test_gpt2 and executed with OMP_NUM_THREADS=8. It passed the logits, initial loss, gradient comparisons, and all ten expected training-loss checks, ending with overall okay: 1. An earlier batch-size-1 debug file produced correct tensor comparisons but failed the fixed training-loss expectations; regenerating the reference with the standard batch size 4 resolved the mismatch. The zero-update timing warning nanms only reflected an empty timing list.

Training setting	Value
Compute and precision	One RTX 3080; FP32; compilation disabled
Microbatch and sequence	1 sequence of 64 tokens
Effective batch	256 tokens; 4 gradient accumulation steps
Optimizer updates	40; overfit_single_batch disabled
Learning rate	0.0001; no warmup; default decay fraction 1.0
Validation sampling	Every 10 updates and at completion; 5 batches per check
Output directory	runs/ascon_demo_40

Table III. Settings for the GPU fine-tuning demonstration.

Validation loss was 5.404110 before training and 3.810759 after 40 updates. Intermediate checks after 10, 20, and 30 updates were 4.078157, 3.942588, and 3.860151. Each value averages five validation batches rather than the entire validation split. The trainer reported a mean of 132.058 ms over its final 19 recorded iteration timings and peak PyTorch allocated GPU memory of 2,416 MiB. These iteration timings are not the total elapsed run time and include any validation work scheduled in a timed iteration.

The supplied PyTorch bridge exports initialization weights before its training loop. An additional export block was therefore added after the loop to call write_model(raw_model, ..., dtype="float32") and save the updated weights as gpt2_trained_fp32.bin. The file size was 497,904,640 bytes: a 1,024-byte header plus 497,903,616 bytes of FP32 parameter data, including vocabulary padding from 50,257 to 50,304 entries. This is approximately 474.84 MiB. The serialized payload contains 124,475,904 FP32 slots; the unpadded model has 124,439,808 unique parameters.

A 1 KiB sample was extracted by skipping the checkpoint header and reading the next 1,024 bytes, equivalent to 256 FP32 values from the first token embedding. The file was verified against that trained payload, and the trained checkpoint was confirmed to differ from the pretrained checkpoint. The sample was copied to Windows as gpt2_trained_weights_1KiB.bin. It represents only a small portion of model weights and cannot reconstruct the model on its own.

IX. Ascon encryption of trained weight bytes

The original official-vector TextIO testbench stores only two 16-byte payload blocks, so it cannot accept the entire 1 KiB sample. A separate runner, verification/gpt2_weights_demo.py, generated a dedicated tb_gpt2_weights testbench and an isolated Vivado project under build/gpt2_weights_demo. The runner reused the existing ascon_pkg.vhd, ascon_permutation.vhd, and ascon_aead128_core.vhd. The application test streamed all 64 full 16-byte beats as one AEAD message; the original RTL and official-vector testbench were not altered.

The Python Ascon reference first passed all 1,089 pinned official known-answer vectors. It then computed the expected ciphertext and tag for the actual trained-weight sample. The testbench checked every ciphertext block and the tag, then decrypted the ciphertext and checked all recovered bytes and successful authentication. Full block dumps were preserved in full_output.txt, while simulation_results.txt and simulation_vectors.txt recorded the PASS results and cycle counts. The runner verified the complete byte arrays before producing report.json.

Measurement or input	Recorded result
Plaintext and associated data	1,024 weight bytes; 0 associated-data bytes
Demo key	000102030405060708090a0b0c0d0e0f
Demo nonce	101112131415161718191a1b1c1d1e1f
Clock	100 MHz; 10 ns per cycle
Encryption	801 command-to-done cycles; 8.01 microseconds; PASS
Decryption	801 command-to-done cycles; 8.01 microseconds; PASS
Payload throughput	1,022.72 Mb/s calculated from simulated latency
Authentication tag	e3a4ec0147a1f6681ba8f8aeb188a49f
Ciphertext and tag sizes	1,024-byte ciphertext and a separate 16-byte tag
Round-trip integrity	Recovered 1,024 bytes exactly match the input sample

Table IV. Behavioral XSim results for one 1 KiB trained-weight sample.

The cycle counter was sampled when a ready command was offered and again when done was observed. Thus the interval includes command submission, initialization, the empty associated-data phase, all payload handshakes, finalization, tag handling, and completion. The driver accepted each output beat before submitting the next input beat. The measurement uses that testbench schedule and must not be described as an ideal unstalled datapath limit. The latency calculation is 801 cycles times 10 ns per cycle, giving 8,010 ns or 8.01 microseconds. Payload throughput is 1,024 times 8 bits divided by 8.01 microseconds.

The source sample SHA-256 is recorded below. The Windows copy matched the Ubuntu export, binding the encryption result to the actual trained-weight bytes. The demo key and nonce are public test inputs. A deployed encryption system would require protected keys and unique nonces for each message under a given key; repeating this fixed pair is appropriate only for reproducing this test.

```
fa00e0b873c026e9c25298fc1428872a0968
0e11e0ec5ac74af29fcab3e34541
```

The 8.01-microsecond result is functional RTL simulation at a chosen 100 MHz clock. No physical board encryption, UART/USB transfer, DMA, PC file I/O, or post-route application timing was measured. The small-payload result of 352.441 Mb/s in Section VI averages different associated-data lengths, whereas this case has empty associated data and a much larger payload. The numbers therefore describe different workloads and should not be treated as a controlled architectural speedup.

X. Historical latency study at a fixed clock frequency

This section preserves the initial architecture proposals and conditional estimates. Section XIII reports the later implemented three-round core, buffering, routed timing and complete adapter simulation. Statements about unimplemented options below describe the earlier study stage.

The initial 1 KiB baseline measured 801 cycles, or 8.01 microseconds at 100 MHz. Published two- and four-round configurations [11] and OpenTitan's latency discussion [12] motivated the proposals below. These sources establish the technique; Section XIII supplies this project's measured follow-up.

A. Round unrolling and conditional cycle estimates

The current permutation engine performs one round per cycle. With no associated data, the 64 full message blocks require 64 eight-round permutations, in addition to twelve-round initialization and finalization. This accounts for $12 + 64 \times 8 + 12 = 536$ round-computation cycles. The remaining 265 cycles include controller transitions, input/output handshakes, and completion work. They cannot all be assumed removable.

Round unrolling places consecutive rounds in the combinational path between state registers. Two rounds per cycle would reduce the round-computation portion to 268 cycles; four would reduce it to 134. Holding the remaining 265 cycles unchanged gives the estimates below. Each estimate assumes that the larger combinational circuit still meets the 10 ns clock period and that controller behavior is otherwise unchanged.

Architecture at 100 MHz	Total cycles	Latency	Reduction
Current one-round core: measured in XSim	801	8.01 µs	Baseline
Two rounds per cycle: conditional estimate	533	5.33 µs	33.5%
Four rounds per cycle: conditional estimate	399	3.99 µs	50.2%

Conditional architecture estimates for the same 1 KiB message and empty associated data. Only the baseline is a simulation result; optimized timing and resource use remain unverified.

Two rounds was proposed first to limit combinational depth. Unrolling increases logic and routing demands, so these cycle estimates require routed timing validation and are not published benchmark measurements.

B. Other ways to reduce waiting

Option	Expected benefit and practical limit
Overlap output transfer and permutation	A registered output buffer can retain the completed ciphertext while the next permutation runs. The current controller waits for output acceptance before starting that permutation. Benefit depends on stalls; ready/valid behavior must remain correct.
Reduce controller handoff cycles	Inspect the start, wait, and completion transitions for avoidable idle cycles. A cycle trace is needed before assigning a numerical saving.
Multiple cores for independent chunks	Separate authenticated chunks can run concurrently, improving completion time for large files. Each chunk needs a unique nonce and tag, with chunk order and model identity authenticated. Transfer bandwidth can limit the benefit.
Pipeline registers within a round	Can shorten the combinational path, but at a fixed clock may add single-message cycles. Consecutive rounds and message blocks depend on the previous state; independent messages are needed to exploit interleaving.

The existing 128-bit input interface already matches the Ascon-AEAD128 message rate. Widening the external bus alone does not remove the sequential permutation dependency. The standard round counts must remain unchanged; the optimization is to perform the required rounds more efficiently, not to reduce cryptographic work.

C. Proposed comparison with the baseline

The proposed comparison was to repeat official-vector and 1 KiB tests after unrolling, recording cycles, routed timing and resources separately from physical-board results. Section XIII now reports that follow-up, including buffering and complete adapter recovery.

XI. Ascon behavior on trained weight bytes

A follow-up experiment on October 6, 2026 tested how the existing Ascon VHDL core responds to changed authenticated inputs, delayed handshakes, and a valid command after authentication failures. All 17 checks across 12 scenarios passed. The input was the same 1,024-byte trained GPT-2 weight sample from Section IX. Five valid scenarios each tested encryption and decryption; seven modified-input scenarios tested rejection during decryption.

The runner verification/gpt2_behavior.py created a separate Vivado 2023.2 XSim project without modifying the RTL or the original testbench. Before generating expected outputs, its Python oracle passed all 1,089 pinned official known-answer vectors. The new testbench compared all 64 payload blocks, encryption tags, authentication events, commit and discard signals, and completion cleanup. Decryption outputs in failed cases were checked against the oracle as provisional plaintext, rather than accepted model weights.

Scenario	Observed behavior	Cycles
Original sample, empty AD	Exact encryption and recovery; authentication passed	801 each
Ciphertext bit flip at byte 0, 512, or 1,023	All three rejected; no commit; discard asserted	801 each
Tag bit flip	Rejected; no commit; discard asserted	801
Wrong key or wrong nonce	Both rejected; no commit; discard asserted	801 each
Valid command after failed commands	Correct encryption and authenticated recovery	801 each
Valid 16-byte associated data	Correct encryption and authenticated recovery	821 each
Associated-data bit flip	Rejected; no commit; discard asserted	821
Input gaps, output stalls, delayed tag	Correct data; held outputs stable; no early authentication	1,003 each
Valid changed nonce	Different ciphertext and tag; correct authenticated recovery	801 each

Table V. XSim behavior results for the 1 KiB trained-weight sample. Each denotes encryption and decryption where both were tested.

The baseline latency remained 801 cycles, or 8.01 microseconds at the simulated 100 MHz clock. Adding 16 bytes of associated data increased it to 821 cycles, or 8.21 microseconds. The stall scenario inserted three-cycle gaps before input beats, held each output beat for three cycles, and delayed tag acceptance by seven cycles. It completed in 1,003 cycles, or 10.03 microseconds. Payload and tag values remained stable while held. These are command-to-done simulation intervals under the specified driver schedule; no physical FPGA or host-transfer time was measured.

Every modified-input case produced auth_ok=0, commit=0, and discard=1. A valid transaction immediately after the failures succeeded. The core still streams provisional plaintext before checking the tag, so an external receiver must quarantine the entire transaction until commit and discard it on failure. This experiment verifies the core signals, not a production authenticated checkpoint loader. Equal valid and invalid latencies in the tested cases do not establish protection against timing, power, or electromagnetic side channels.

XII. GPT-2 behavior after weight recovery

The full GPT-2 model produced identical results when its tested weight slice was replaced with the newly recovered RTL bytes. This experiment used the complete 497,904,640-byte trained FP32 checkpoint, but only the first 1 KiB of parameter data passed through Ascon RTL. That slice contains 256 FP32 values from token embedding row 0; it begins immediately after the checkpoint header. Whole-checkpoint encryption was not performed.

The runner verification/gpt2_model_behavior.py loaded two model instances using the actual checkpoint tensor order and padded vocabulary format. One instance used the original checkpoint bytes; the second substituted the recovered slice. Every model parameter matched. Reserializing both instances reproduced the original full-checkpoint SHA-256, c5acb338337d9f14b814a21e1d18415ce72c34bf18650c9ead5299b4f3ad2733. This check binds the loaded models to the trained export and guards against a wrong tensor order or an unintended model source.

Inference ran in evaluation mode on the NVIDIA RTX 3080 Laptop GPU using PyTorch 2.11.0+cu128. FP32 arithmetic, deterministic operations, and disabled TF32 controlled the numerical comparison. Five text prompts and an explicit token sequence [0, 1, 2, 3] were evaluated. Token 0 directly exercises the embedding row containing the recovered values. Each text prompt also generated 24 new tokens using greedy decoding and a separate seeded top-40 sampling run at temperature 0.8 with seed 20261006.

Comparison	Measured result
Parameters and serialized checkpoint hash	Exactly identical
Prompt logits across six input sequences	1,306,682 FP32 values; all exactly identical
Maximum absolute prompt-logit difference	0.0
Greedy generation	Five prompts; 24 new tokens each; identical token sequences
Seeded generation	Five prompts; 24 new tokens each; identical token sequences
Held-out validation sample	20 batches of 64 predicted tokens; 1,280 tokens total
Mean cross-entropy with original weights	4.0104284048
Mean cross-entropy with recovered slice	4.0104284048
Validation logits and per-batch losses	Exactly identical in all 20 batches

Table VI. Full-model inference comparison with the original and RTL-recovered 1 KiB weight slice.

Validation used the first 20 non-overlapping 64-token input batches from the held-out Tiny Shakespeare token file, with next-token targets. Its mean cross-entropy was 4.0104284048 for both model instances. Section VIII reports a different five-batch evaluation mean of 3.810759 after training; the different sampling schedules prevent a direct before-and-after quality comparison between those two means. The behavior test compares original and restored weights on the same inputs and obtains zero change.

Together, these results demonstrate exact recovery of the tested weight bytes, rejection of the seven tested modifications, and preservation of full-model inference behavior after restoring that slice. They do not establish model quality, whole-checkpoint RTL encryption, or a physical-board application. The public evidence folder paper/evidence/gpt2_behavior_20261006 preserves the runner and testbench snapshots, compact simulator results, and machine-readable report.json and model_report.json, including generated text and token sequences. Trained weight binaries, full block dumps, and generated Vivado projects remain local.

XIII. Buffered three round adapter results

The complete SAGE adapter export was encrypted and recovered with half the baseline cycle count at a 100 MHz simulated clock. The unchanged one-round core required 1,798,305 cycles per direction; three rounds per cycle with input/output buffering and controller overlap required 899,153. This approximately doubles throughput and halves the cycle count. Both runs recovered every byte and passed RTL authentication and commit. The result measures encryption and decryption throughput, not model inference speed.

A. Architecture and sources

NIST SP 800-232 defines the required Ascon-AEAD128 behavior [1]. The implementation retains its 128-bit rate, padding, round constants, twelve-round initialization/finalization, and eight-round block permutations. A configurable combinational chain defaults to one round per clock. The candidate evaluates p8 as 3 + 3 + 2 rounds and p12 as four groups of three. The last p8 group executes only its two remaining rounds; it neither omits a round nor executes a ninth round.

Unrolling reduces round-work clocks at the cost of a deeper combinational path. For n rounds and r rounds per clock, the round work takes ceil(n/r) clocks; loading and controller handoffs are additional. Each round consumes the preceding round's complete state, so a single permutation uses cascaded rounds rather than independent parallel rounds. Fig. 4 shows the three-round chain and its groups: p8 uses 3+3+2 and p12 uses 3+3+3+3. The last p8 group bypasses lane 2. In the saved buffered candidate, the worst setup path passes through this chain and its load multiplexer: seven LUT levels and 9.909 ns of data delay at a 10 ns clock [14].

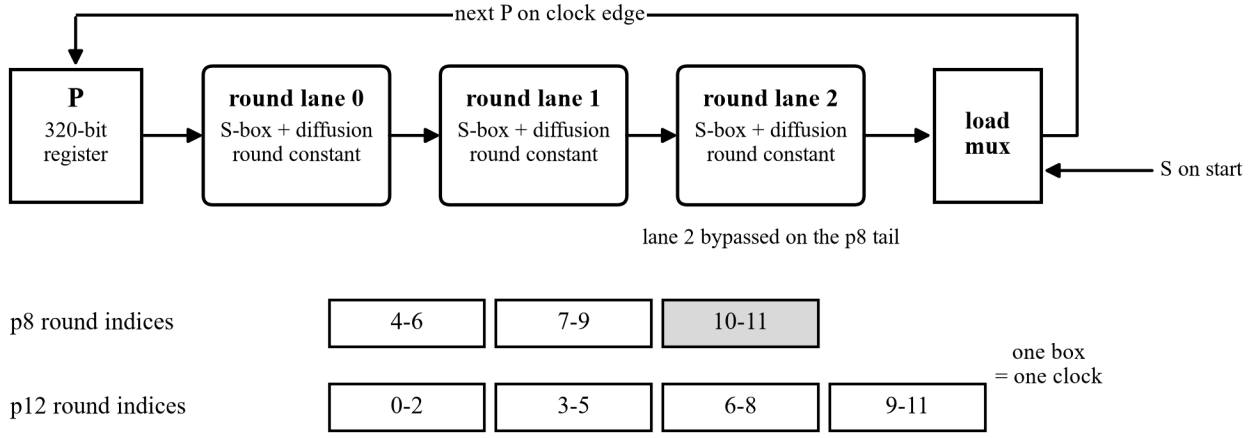


Fig. 4. Three-round permutation chain. Each lower box is one clock, with zero-based round indices. Lane 2 is bypassed in the gray p8 tail, preserving exactly eight rounds. The load and completion-copy clocks are not included in these round groups.

The earlier architecture study drew on Robert Primas's public Ascon hardware variants, which provide one-, two-, and four-round unrolling [11], and OpenTitan's discussion of latency, feedback dependencies, and the area cost of unrolling [12]. Those sources motivate the architectural technique; they do not supply this project's three-round performance numbers. OpenTitan also limits its unrolling rationale to an unhardened datapath. This core remains unmasked and makes no side-channel-resistance claim.

The two-entry input FIFO accepts beats while the controller is busy, and the output FIFO accepts a completed beat while the consumer is stalled (Fig. 5). They retain data and kind/keep/last metadata in order. Transfers follow ready/valid: unaccepted data stays stable and interface state advances only on a handshake [13]. The controller can launch p8 on the edge that enqueues an output beat, instead of waiting for that beat to reach the consumer. This merges output transfer and launch, reducing the saturated full-block interval from seven clocks to six. The controller and permutation still own separate state registers. Buffering hides some source and sink gaps; it does not remove every possible stall.

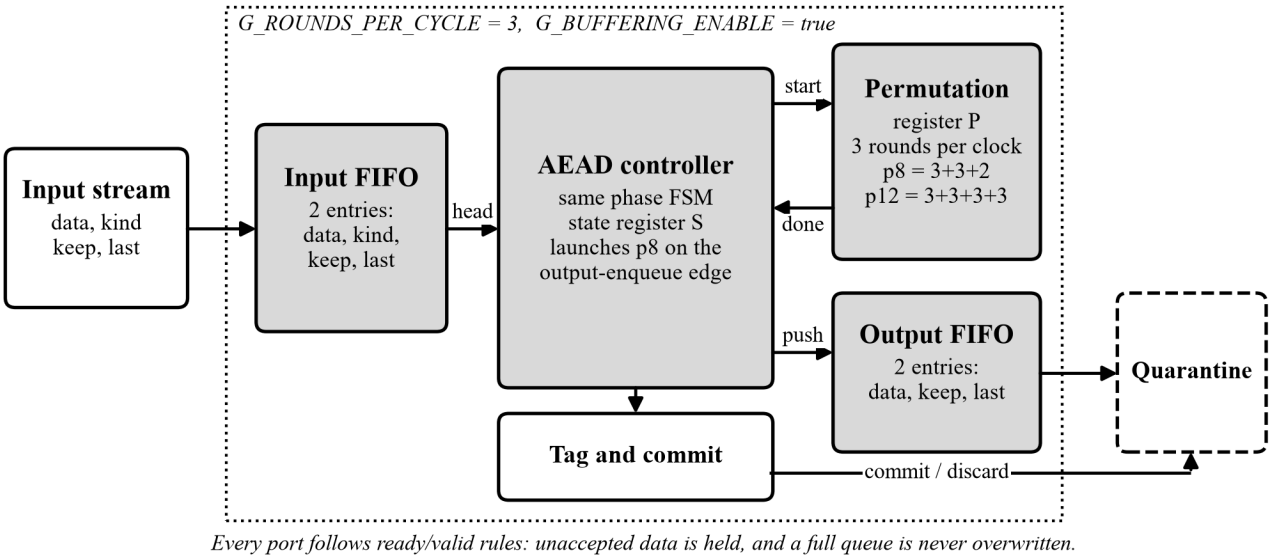


Fig. 5. Buffered three-round core. Gray blocks show the changes from Fig. 2: input/output FIFOs, the three-round permutation, and output-enqueue launch overlap. S and P remain separate. Command, key and tag-input wiring is omitted; tag decisions still come from the controller.

Decrypted bytes remain provisional. An external consumer must collect them into quarantine while the stream runs and publish only on commit_o; discard, protocol error, or reset invalidates that transaction. Authentication is delayed until queued plaintext drains, and pending ciphertext is retained through completion. Reset and protocol errors erase queued data and cryptographic storage. A selectable full-16-byte mask specialization saved no clocks and increased area in the controlled experiment, so it is disabled. Partial endings and mandatory final padding remain supported.

B. Adapter identity and simulation method

The existing file runs/adapter_fp32_200/export/adapter_weights.bin contains all 599,424 little-endian FP32 adapter values: down_w [12,32,768], down_b [12,32], up_w [12,768,32], and up_b [12,768]. Its length is 2,397,696 bytes (149,856 full blocks), with no export padding. The saved checkpoint is adapter_params_00000200.bin. The export is exactly its complete body after the 20-byte [B,L,C,r,T] format header. This experiment used saved artifacts and did not train a model.

On October 20, 2026, Vivado 2023.2 XSim ran the buffered three-round core with a 10 ns behavioral clock. A fresh GHDL run of unchanged origin/main reproduced the earlier one-round result exactly on the same file. Both file benches count command through done, including stream handshakes, empty associated data, mandatory final padding, and tag work; host I/O, Python reference execution, and simulator wall time are excluded. Output is always ready in this throughput test. Independent encryption runs use distinct nonces under a public test key. The corresponding decryption reuses each message’s nonce as required to verify it.

Configuration at 100 MHz	Cycles each	Latency each	Payload MB/s
Unchanged one-round baseline	1,798,305	17.98305 ms	133.331
Three rounds with queues and overlap	899,153	8.99153 ms	266.662

Table VII. Complete adapter payload; encryption and decryption have equal latency. MB denotes 1,000,000 bytes. The speedup combines unrolling, buffering, and launch overlap.

The Python oracle first passed all 1,089 official known-answer vectors. The full-file RTL runs then matched every ciphertext block and tag against that oracle, asserted authentication and commit, and recovered all payload bytes with zero mismatches. Original and recovered payload SHA-256: c184175fc18d24373f009875c03698e630e8c36a5341146aab2bd13a98fff8eb. Reattaching the original format header reproduced the complete saved checkpoint exactly, SHA-256 7c45c25b28ae54943c1fd4ee160001dcd9a68ee1f8430aee229f933545672de6. The header itself is outside the requested exported weight payload and was preserved unchanged.

C. Correctness coverage and comparison with earlier workloads

All three core configurations passed all 1,089 official vectors in both directions, or 2,178 checks per configuration. Additional tests covered concurrent sustained traffic, input gaps, long output stalls, FIFO capacities one through three, wraparound and simultaneous enqueue/dequeue, reset with buffered data, malformed streams, consecutive messages, empty

inputs, all partial final lengths, and tampered ciphertext, associated data, nonces, and tags. Rejected transactions must never commit. Intermediate permutation-state traces check grouping without changing cryptographic round counts. These protocol checks are separate from the always-ready adapter throughput run.

Configuration	1 KiB cycles	1,641,984-byte cycles	Full-block interval
One round	801	1,231,521	12 clocks
Three rounds alone	465	718,385	7 clocks
Three rounds with queues and overlap	401	615,761	6 clocks

Table VIII. Separate complete small-LLaMA payload comparison from October 20, 2026. These are measured simulation counts, not estimates for the adapter file.

The adapter comparison does not separately measure three-round unrolling alone. Table VIII isolates that stage on another complete payload. Under a 1 KiB schedule with associated data, input gaps and output stalls, buffering reduced three-round cycles from 862 to 581. With always-ready traffic it reduced block spacing from seven clocks to six. Later command-relative traces isolate one clock of FIFO first-output latency; the earlier two-clock difference also includes a command-boundary wait [14]. Three clocks of the remaining six execute p8 round groups; request, completion and block-handling overhead remain. This explains why tripling round execution per clock gives approximately twice the complete-file throughput.

D. Routed timing and physical evidence

Core-only configuration	Setup / hold ns	LUTs	Registers
One round	+2.914 / +0.147	1,894	1,081
Three rounds alone	+0.125 / +0.152	3,192	1,088
Three rounds with queues and overlap	+0.038 / +0.144	3,708	1,680

Table IX. Vivado routed core-only results for xc7a100tcsg324-1 with a 10 ns clock. External I/O paths were unconstrained; these are internal core timing results.

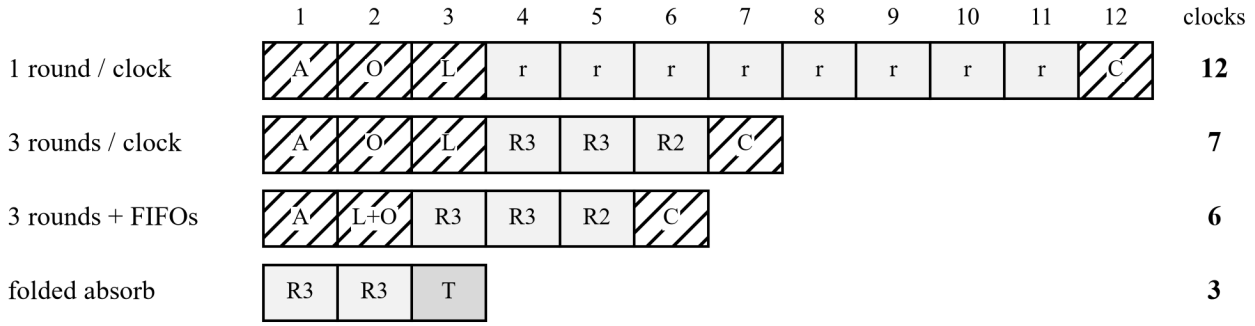
The final board-integrated design contains separate demo and self-test cores plus ILA. It closed at 83.333 MHz with +1.110 ns setup and +0.024 ns hold slack, using 9,503 LUTs and 6,510 registers. Its MMCM generates a 12 ns clock from the board’s 100 MHz reference. An earlier board trial at 100 MHz failed setup at -0.130 ns; it predates the final demo fix and is not a timing result for the final source. The core-only 38 ps setup margin does not establish 100 MHz board closure.

Physical JTAG ILA captures on the confirmed Nexys A7-100T show completion/pass with no failure for 18 core self-test cases and eight demo vector/mode pairs. Latched counts were 1,161 and 274 cycles respectively, matching simulation. These captured bounded fixtures establish more than programming success, but do not establish full adapter transport or physical weight throughput. Complete adapter recovery remains simulation evidence. Full-file board testing would require a host transport beyond this core.

E. Reproduction and remaining bottlenecks

The new verification/simulate_adapter.py runner validates the export manifest and optional original checkpoint, then freezes unchanged origin/main and buffered candidate sources and runs the 1 KiB prefix and complete file at 10 ns. It checks all output/tag bytes and authenticated recovery, publishes recovered files only after PASS, reconstructs the checkpoint when supplied, and records source hashes, nonces and cycle counts. paper/evidence/adapter_100mhz_20261021 contains compact reports, manifests and simulation logs. docs/evidence/weight_stream_3rpc contains the wider protocol, implementation and hardware evidence. Binary weights, full dumps, simulator projects and bitstreams remain local.

Fig. 6 separates round work from controller overhead. In the one-round design, eight clocks execute p8 and four handle absorption (A), output acceptance (O), permutation loading (L) and completion copying (C). Three-round unrolling changes only the round term: 4 + ceil(8/3) = 7 clocks per full block. Complete-adapter traces record the same 599,433 non-round clocks for the unbuffered one- and three-round runs [14]. This is the fixed-overhead limit described by Amdahl’s law. Buffered output/launch overlap merges O and L to reach six clocks; three clocks still perform no round work. Section XIV removes the second state owner and folds the next absorption into the short tail, producing the three-clock row.



Hatched: no round work. A absorb (XOR, mask, pad); O output handshake; L launch (copy S into P); C completion (copy P back); r one round; R3/R2 three or two rounds; T two tail rounds plus absorption of the next block. Measured with always-ready traffic.

Fig. 6. Saturated full-block intervals in simulation: 12, 7, 6 and 3 clocks. Hatching marks clocks without round work. A: absorption; O: output acceptance; L: load S into P; C: copy P back; r: one round; R3/R2: three/two rounds; T: two tail rounds plus the next absorption. Rows show recurrence intervals, not command latency or aligned processing of the same block.

These counts describe one dependent message with available input and output space. Independent-message lanes require separate contexts and arbitration. Physical memory transport needs DMA or bus adaptation and a memory controller. Lossless compression adds framing, metadata and a decoder whose throughput must be measured on these bytes. Those system costs are outside the buffered-core measurements; Section XIV evaluates the implemented folded datapath.

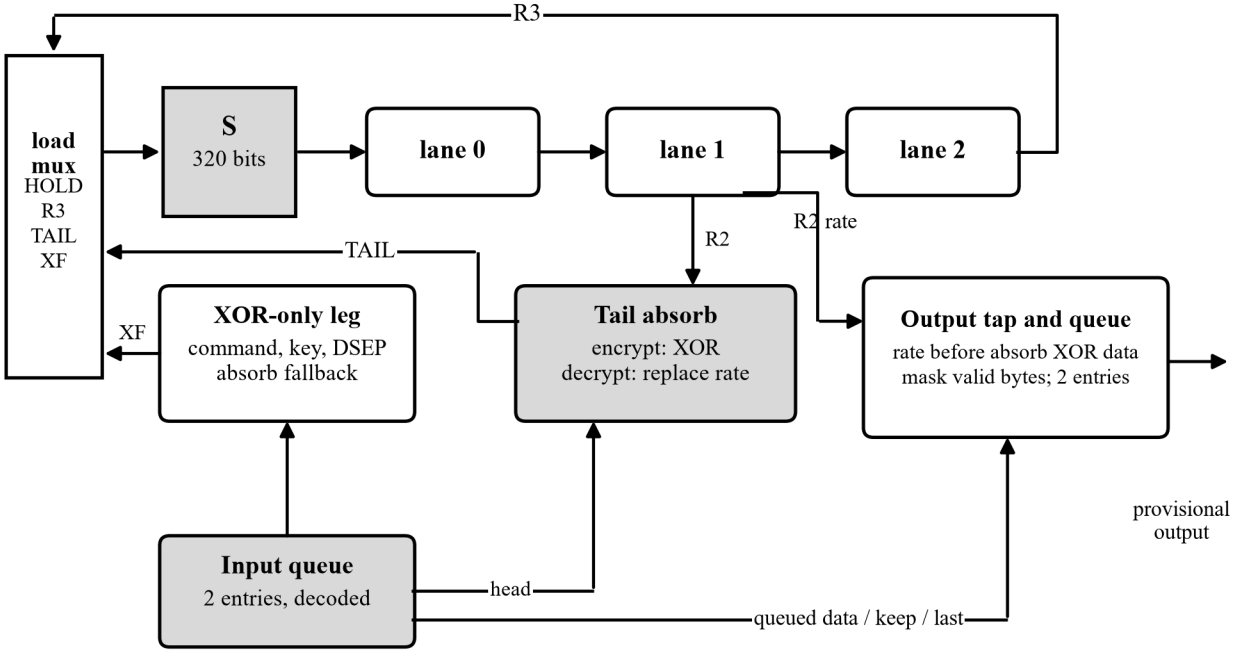
XIV. Folded absorption and controller fixes

The folded core reduces the complete adapter run to 449,581 cycles per direction, compared with 899,153 for the buffered three-round core and 1,798,305 for the original one-round baseline. At the simulated 100 MHz clock, this is 4.49581 ms and 533.318 MB/s. Encryption matches the reference ciphertext and tag; decryption authenticates, commits and recovers all 2,397,696 bytes. Reattaching the preserved 20-byte header reconstructs the saved checkpoint exactly. These are GHDL simulation results, not measured memory or host-transfer rates [14].

A. State ownership and the three clock schedule

G_FOLDED_ABSORB selects a separate entity with one 320-bit state register S. It requires G_ROUNDS_PER_CYCLE=3 and G_BUFFERING_ENABLE=true and is disabled by default. S feeds the round chain directly. Its load multiplexer selects hold, the three-round result (R3), the two-round tail with absorption (TAIL), or an XOR-only leg (XF) for command loading, key and domain-separation terms, and absorption without rounds (Fig. 7). There is no second state register or registered start/done interface. The load and completion-copy clocks in Fig. 6 therefore disappear.

Absorption then uses the spare depth in the 3+3+2 schedule. After the final two p8 rounds, encryption XORs the queued plaintext into the rate and emits ciphertext. Decryption emits the old rate XOR the queued ciphertext, then replaces the valid rate bytes with ciphertext. The next p8 group runs on the following edge. The resulting recurrence is R3, R3, TAIL; p12 remains four groups of three, with no omitted rounds. In the saved route, the state-to-output-queue tap had +1.373 ns slack, while the worst state-update path remained seven LUT levels and 9.841 ns [14]. That output-tap result is not a separate measurement of every path through the tail. A 4+4 grouping has no short group and would place absorption after four rounds.



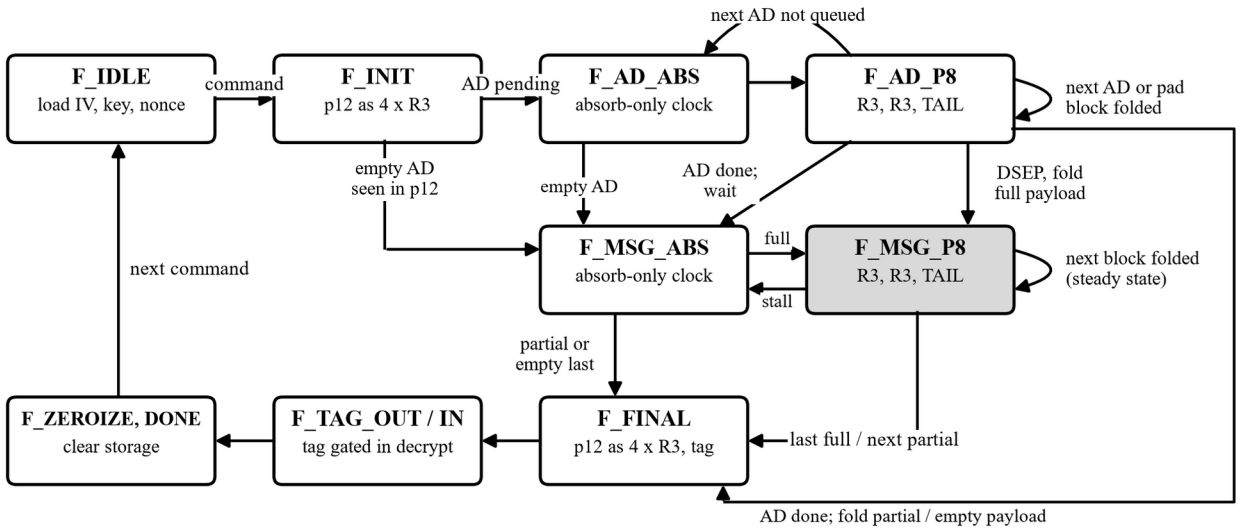
Output fallback selects the rate from S. Tag, command and controller wiring are omitted.

Fig. 7. Folded datapath with one state owner. Gray blocks show S, the decoded input queue and tail absorption. The output tap uses the rate before absorption, from S or the two-round result, XORed with the queued data. Tag and external command wiring are omitted. The multiplexer selects HOLD, R3, TAIL or XF.

Steady state edge	State update
1	First three p8 rounds
2	Next three p8 rounds
3	Final two p8 rounds and next block absorption

Table X. Saturated full-block schedule. Initialization, finalization and stall recovery are separate.

Input beats are decoded on queue acceptance, preparing masked data, padding and error flags before the tail edge. The queue head stays at entry zero, and round constants are registered one clock before use. Input ready is derived from registered phase and queue state rather than from input valid or downstream ready; reset still gates the interface combinationally. The FSM in Fig. 8 uses absorb-only states for the first beat and whenever a tail cannot absorb the next beat. It finishes p8 without absorption if input is absent or output space is unavailable, then waits for an eligible beat. Final full, partial and empty blocks retain their separate padding rules [1], [13]–[15].



Absorb-only states handle first beats and tail fallback. Group advances, waits and errors are omitted; errors enter F_ZEROIZE.

Fig. 8. Grouped folded-controller flow. The gray state sustains one full block per three clocks when input and output space are available. Absorb-only states handle initial absorption and tail fallback. Partial/empty final payloads can enter F_FINAL directly from either absorption path. p8/p12 group advances, waits and error edges are omitted; protocol errors enter F_ZEROIZE.

Other hardware reports need separate interpretation. Liu, Han and Quan describe a larger parallel architecture using legacy Ascon-128 [4]; their throughput is not a single-stream SP 800-232 result. Huong, Nam and Phat report an IoT application of Ascon-AEAD128 [5]. Neither source supplies performance measurements for this adapter or board wrapper.

B. Verification and saved implementation results

Configuration	Full adapter cycles	1 KiB cycles
Original one-round baseline	1,798,305	801
Buffered three-round core	899,153	401
Folded three-round core	449,581	205

Table XI. Command through done with empty AD and always-ready output. Counts are equal for encryption and decryption [14].

The folded core passed 2,178 official directional checks, concurrent stream tests, partial blocks, stalls, reset and malformed-input checks. test_fold compares intermediate round groups with the Python model and checks a mid-stream error with queued plaintext. test_cycles now checks documented cycle counts. Generated-Verilog smoke tests passed with Icarus and Verilator before the later legacy-engine fixes. The final tag-gate evidence records twelve test_ascon groups passing for all four configurations, full KAT runs for u1, buffered u3 and fold, full stream suites for buffered u3 and fold, and cycle regressions. Restoring the ungated legacy assignment makes the new tag-output test fail, as expected [14], [16].

Core-only snapshot at 100 MHz	Setup / hold ns	LUT / FF
Buffered u3 before both fixes	+0.038 / +0.144	3,708 / 1,680
Buffered u3 after AD fix, before tag gate	+0.287 / +0.110	3,871 / 1,688
Folded core	+0.139 / +0.101	2,985 / 1,085

Table XII. Vivado 2023.2 routes on xc7a100tcsg324-1. External I/O delays are unconstrained; the final legacy tag gate has not been rerouted [14], [16].

Relative to the earlier buffered snapshot, the fold saves 723 LUTs and 595 registers. Its setup path still has seven LUT levels, with 9.841 ns data delay. The small slack differences are not evidence of a higher maximum frequency. AMD notes that 7-series LUT delay is included in net delay, so a large reported routing fraction cannot be interpreted as removable wire delay alone [17].

In separate RTL copies with both board cores folded, the board wrapper routed at 83.333 MHz with +0.785 ns setup and +0.024 ns hold slack, and at 100 MHz in one run with +0.181 ns setup and +0.049 ns hold slack. The 100 MHz build used 9,690 LUTs and 5,362 registers. Board LUT savings did not follow the core-only reduction. At this October 21 stage, no folded bitstream was written or programmed; the later multicore fixture is reported in Section XVI. The earlier physical captures belong to the buffered core; fold board simulations passed the self-test in 995 cycles and all eight demo pairs in 198 cycles. Full-adapter physical streaming remains unmeasured [14].

C. Associated data framing and tag output fixes

The legacy engine incorrectly treated an empty last AD beat as empty AD even when full non-last AD beats preceded it. It skipped the padded block and p8, producing non-standard output and allowing two distinct AD strings to reach the same authentication state. The fix records whether any non-empty AD has been absorbed. An empty terminator after non-empty AD now takes the existing padding path; genuinely empty AD still skips p8. The folded entity already handled this case. Official-vector framing and the empty-AD adapter workload were unaffected [16].

A second fix holds tag_out_o at zero during legacy decryption. The previous assignment exposed the expected tag while the core awaited tag_in_i; an observer could read the correct tag for modified ciphertext and return it on tag_in_i, causing that message to authenticate and commit. Encryption retains the extra tag-visible clock used by the existing demo. The folded entity already gated its tag output. Default configurations therefore intentionally differ from the published 8b886c0 source: pre-fix netlist-identity evidence is historical, not a claim about the final patched engine. Saved final regressions and negative controls cover the gate [16].

D. Weight loading and the remaining system boundary

The core emits provisional plaintext, so a trusted consumer must accept it into quarantine and release it only on commit_o. Discard, reset or protocol error invalidates that transaction. A FIFO is not a complete-message authentication buffer. Independent authenticated chunks can bound storage and allow loading to overlap compute, but require unique encryption nonces, authenticated model and tensor identity, offsets, lengths, chunk count and a trusted version policy. AEAD alone does not prevent rollback or hide weights from an authorized plaintext endpoint. GuardNN motivates workload-aware memory protection; its AES confidentiality-only FPGA prototype is not an Ascon integrity-performance measurement [18].

The weight study found that lossless exponent-oriented compression reduces this export to about 83% of its original size, with byte-exact recovery. That does not establish a faster integrated loader. A decoder must sustain the folded core and must not expose unauthenticated bytes. The next measurements should test folded-core timing across implementation strategies and actual memory service gaps before adding lanes. A two-round folded schedule is a separate, untested candidate [14].

XV. Dependent p8 round pipelining and controller timing

The isolated one-round experiment found no latency benefit from splitting a dependent round. The best tested design retains the original one-round datapath and optimizes its controller: 12 clocks at a 4.15 ns passing routed period, or 49.8 ns per block, compared with 70.8 ns for baseline A. This is a 1.422 times elapsed-time improvement. These are routed periods multiplied by simulated cycle counts; no pipeline variant was programmed or measured on the FPGA [19].

This experiment starts from main commit 3cceb3c and uses one message context, unchanged nonce semantics and no round unrolling or interleaving. It is separate from the folded and multicore designs. A denotes accept and absorb, O the output handshake, L the permutation load, and C the final capture. The schedule A, O, L, eight p8 rounds with constants B4 through 4B, C gives the measured 12-clock block interval. A two-stage reused round needs 20 clocks and a three-stage reused round needs 28. Initialization and finalization still use p12, with 12, 24 or 36 round clocks respectively.

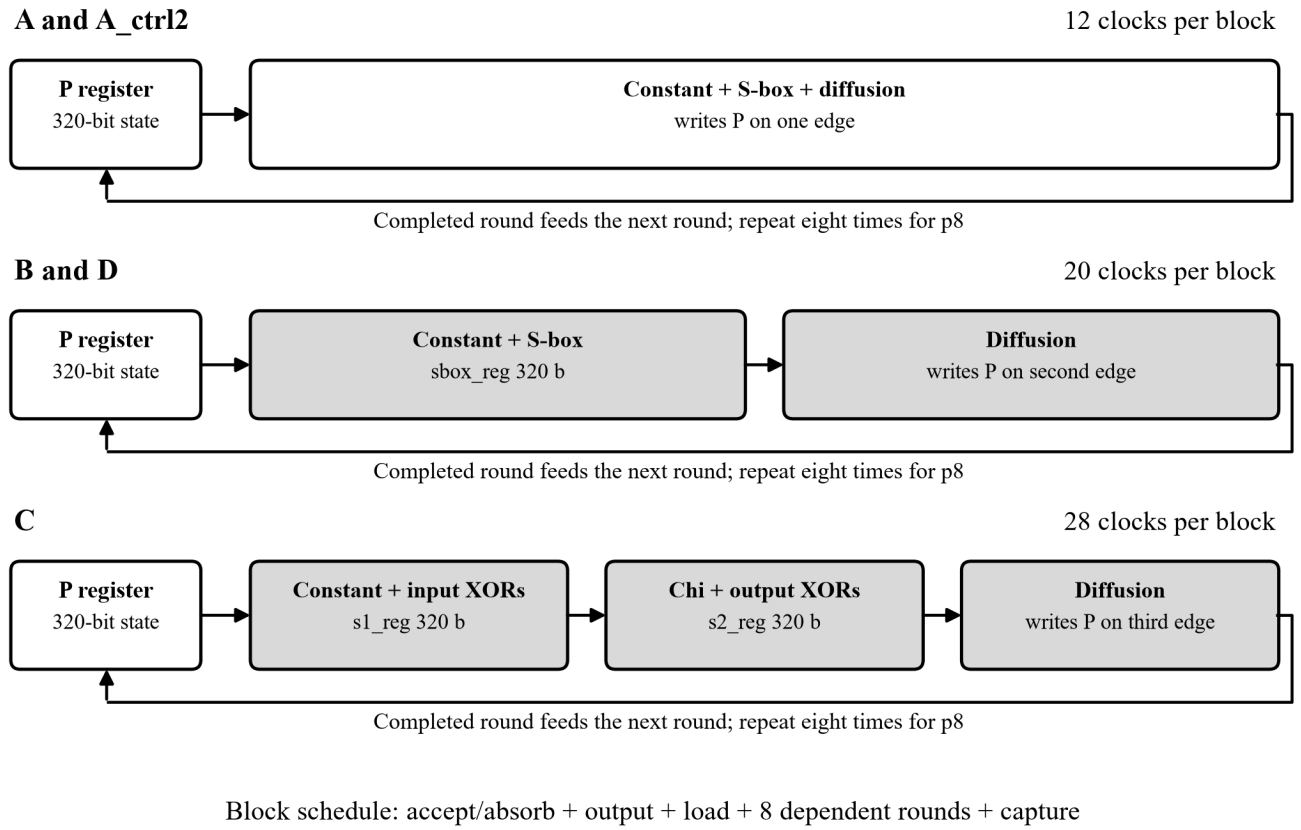


Fig. 9. Implemented round boundaries for A, B and C. Each stage result is registered; the P register holds only completed rounds. Dependent rounds do not overlap. D uses B with controller fixes; A_ctrl2 retains A.

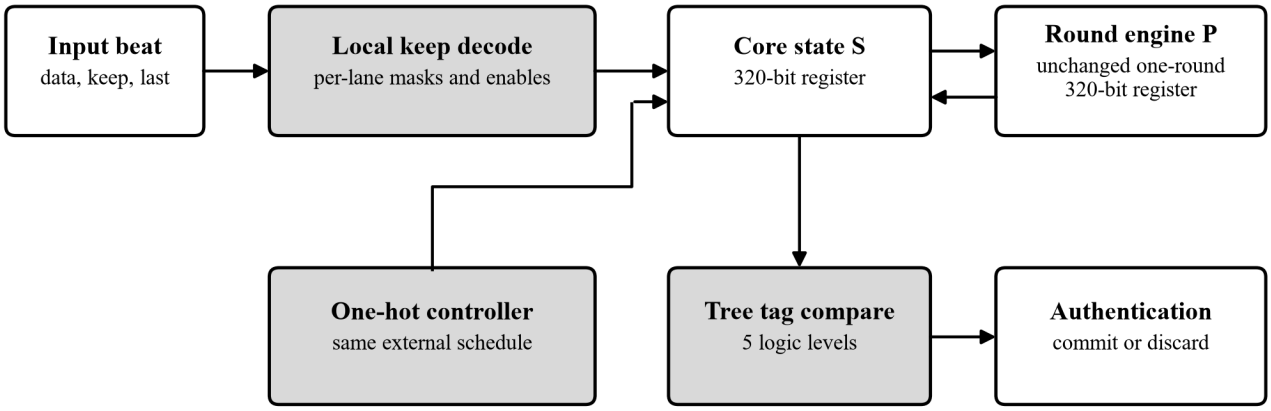
Variant	Cycles	Pass / next fail ns	MHz	Block ns	Speedup
A baseline	12	5.90 / 5.85	169.5	70.8	1.000
B two-stage	20	6.20 / 6.15	161.3	124.0	0.571
C three-stage	28	6.35 / 6.30	157.5	177.8	0.398
D B plus fixes	20	4.30 / 4.25	232.6	86.0	0.823
A_ctrl1 decode	12	4.70 / 4.65	212.8	56.4	1.255
A_ctrl2 all fixes	12	4.15 / 4.10	241.0	49.8	1.422

Table XIII. Best tested passing periods at 0.05 ns resolution. Speedup is 70.8 ns divided by block latency. A, D and A_ctrl2 use explore; the others use default [19].

A. Critical paths and the control experiment

At baseline A's 5.90 ns target, the round has about 1.9 ns of spare time. The limiting path is in_keep decoding into wide state-register enables: six logic levels, approximately 1.2 ns logic and 4.7 ns reported routing, with fanout 129. Round splitting preserves this limit while adding registers and control load. B and C reach only about 0.97 and 0.94 times A's frequency, against required ratios above 20/12 = 1.667 and 28/12 = 2.333. C's stages already take about 2.4 to 2.5 ns each; balancing those stages alone does not remove the limiting control path. Reported routing fractions are Vivado categories and do not imply that all net delay is removable wire delay [17].

D was built on B because its measured latency was lower than C's. Control optimization raises D to 232.6 MHz, or 1.37 times baseline A's frequency, but its 20 clocks still take 86.0 ns. Applying the same changes to A is the decisive control experiment. Per-lane keep decoding removes keep-count and validation logic from wide enables; default-strategy frequency rises from 166.7 to 212.8 MHz. A one-hot controller and a tree-shaped tag compare, five levels instead of thirteen, complete A_ctrl2. The combined changes reach 241.0 MHz at the same 12 clocks. B adds 315 core registers and C adds 638; A_ctrl2 uses 2,271 LUTs and 1,090 registers versus A's 2,369 and 1,080.



A_ctrl1: local decode. A_ctrl2: all three gray changes. D: same fixes with B round engine.

Fig. 10. Controller changes that produce the measured gain. Gray blocks show local keep decoding, one-hot control and the tree tag compare. A_ctrl2 keeps the single-clock round and both original state owners; command, padding and output details are omitted.

Explore improves the best periods by only a few percent: A 6.00 to 5.90 ns, A_ctrl2 4.30 to 4.15 ns, and D 4.50 to 4.30 ns. Retiming never improves these results and more than doubles A's register count. A_ctrl2's remaining limits include tag comparison, controller and done fanout, and the round itself, now only 0.48 ns from critical. A 20-cycle candidate must close every path below 4.15 times 12/20 = 2.49 ns, above 401.6 MHz, to beat A_ctrl2. The 241 MHz result is the highest tested pass, not a measured hardware maximum.

B. Verification and measurement boundaries

Each final variant passes 2,178 official directional checks, 400 per-round permutation runs, 520 unstalled streaming transactions and 520 transactions with random stalls. Tests cover empty, partial and full blocks, 1 KiB and 4 KiB messages, forged tags, and resets throughout the pipeline phases and during transactions. Saved logs also cover D_alt_3stage, which was verified but not timing-swept. Encryption and decryption have identical unstalled cycle counts. With no AD, A_ctrl2 takes 801 clocks or 3,324.2 ns for 1 KiB, and 3,105 clocks or 12,885.8 ns for 4 KiB; these are message simulations, not complete-adapter or memory-transfer measurements.

All twelve reported best implementation runs have nonnegative setup, hold and pulse-width slack, zero routing errors, zero DRC errors and zero DRC critical warnings. No timing exceptions are used. Internal endpoints are constrained; the asynchronous button input and LED output are the only unconstrained I/O. The registered timing harness prevents unused logic from disappearing. Source hashes and comment-stripped pre-relabel hashes preserve measurement provenance. The saved next failing period brackets each reported passing point but does not prove a global frequency limit. Faster board operation requires an MMCM or PLL and reset held until synchronized lock; editing create_clock does not change the oscillator.

XVI. Folded multicore architecture and physical fixture

The folded multicore experiment assigns twelve independently authenticated layer chunks to one, two or four folded cores. In full-adapter simulation, two cores nearly halve the one-core cycle count, while four approach the shared 128-bit input and output bus ceiling. A later FPGA test validates all three core counts on a bounded 3,072-byte adapter fixture. It does not measure transfer of the complete adapter or service from DDR2 [20].

The complete 2,397,696-byte FP32 payload is gathered into twelve 199,808-byte chunks. Each chunk has 96 bytes of authenticated metadata, its own nonce and tag, and preserved chunk/offset identity. Input arbitration admits at most one 128-bit beat per clock across all lanes; the output collector also serves at most one beat per clock. Each lane has an optional registered-ready elastic boundary around a folded 3+3+2 core. Chunks are distributed by lane index and then advanced by the core count. Provisional decrypted output must terminate in trusted quarantine; transaction commit permits release and discard, abort or reset invalidates it. The diagram's quarantine is a required system consumer, not a complete-adapter storage block implemented by this wrapper.

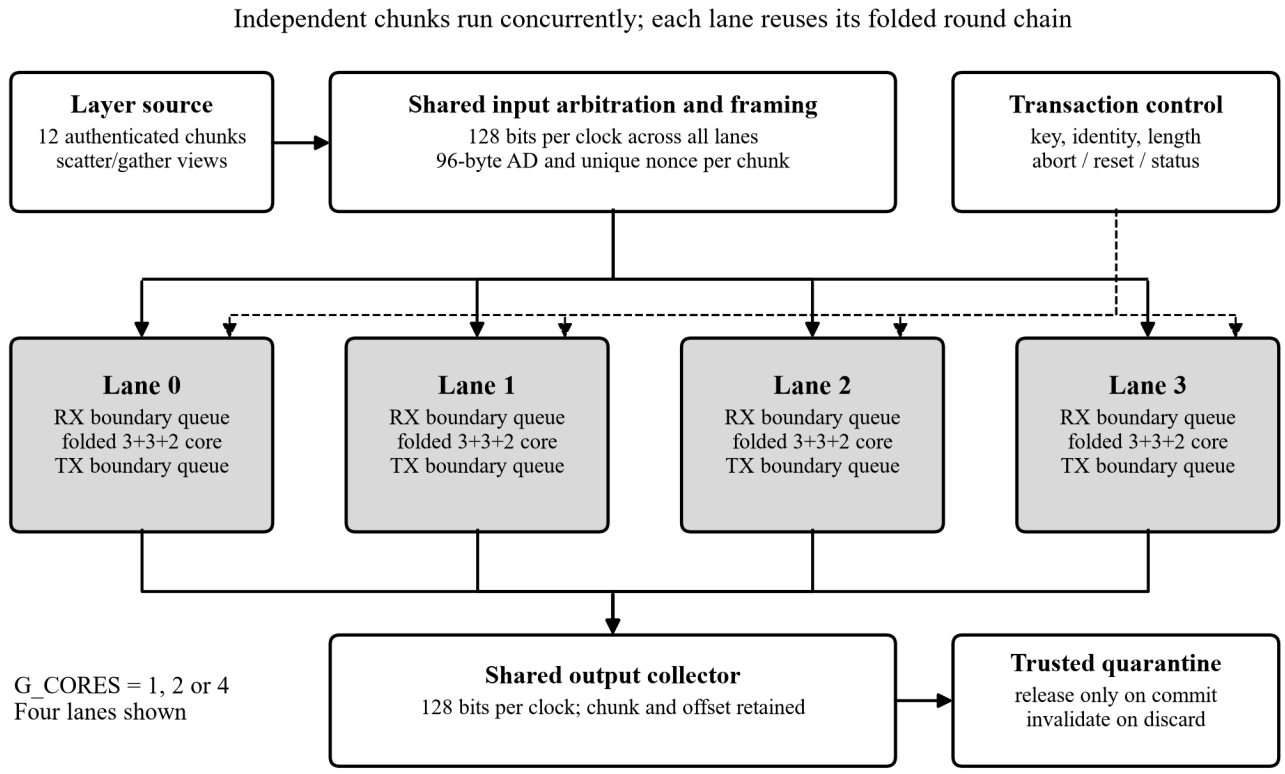


Fig. 11. Four-lane view of the configurable one-, two- and four-core wrapper. Shared input framing and output collection bound aggregate traffic. Each core independently authenticates its chunks; transaction control collects completion and failure. Memory hardware is outside this architecture.

Cores	Encrypt cycles	Decrypt cycles	Encrypt MB/s	Speedup
1	450,013	450,001	532.806	1.000
2	225,008	225,002	1,065.605	2.000
4	149,968	149,965	1,598.805	3.001

Table XIV. Full-adapter always-ready simulation at a nominal 100 MHz, with boundary queues bypassed. Decimal payload MB/s excludes metadata. These are simulated kernel results [20].

A. Bus service and routed integration timing

All reported full-adapter encryptions match every reference ciphertext beat and tag. Decryption reconstructs the file and checkpoint exactly. Registered-ready boundaries preserve similar ideal throughput: 450,013, 225,007 and 149,971 encryption clocks for one, two and four cores. Two folded cores request about 1,066.7 MB/s of payload; four would request about 2,133.3 MB/s without bus contention. Four therefore approach the separate buses' 1,600 MB/s ceiling at 100 MHz. Four-core cycle

speedup is about 3.001 times one chunked core, 5.996 times buffered u3 and 11.991 times original u1; those denominators describe different architectures.

The memory model charges reads and writes, including AD and tags, against an assumed shared budget. At 1 GB/s aggregate service, two and four cores both take about 4.797 ms to encrypt; at 1.3 GB/s they take about 3.749 ms. A payload read plus write already costs 4,795,392 bytes, so ideal lower bounds are 4.795392 and 3.688763 ms. A later release copy is excluded. These are modeled service limits, with no physical DDR2 controller or measured memory bandwidth.

Integration harness	MHz	Setup / hold ns	LUT / registers	Derived encrypt ms
1 core, boundaries	100	+0.004 / +0.098	6,980 / 3,873	4.50013
2 cores, boundaries	100	+0.012 / +0.098	12,776 / 5,908	2.25007
4 cores, local control	90	+0.060 / +0.098	25,916 / 10,966	1.66634

Table XV. Selected routed integration harnesses on xc7a100tcsg324-1, Vivado 2023.2. Time is simulated registered-boundary cycles divided by routed clock. This is not physical file latency [20].

Unregistered base routes fail 100 MHz for all three counts. Boundary registers close one and two cores; four fail at 100 MHz even with local control and Pblocks, then pass at 90 MHz. The one- and two-core critical paths are seven-level folded state feedback; four cores are limited by twelve-level framing/control into the core input boundary. These are individual routes with margins as small as 0.004 ns. Internal register timing is assessed separately from nominal unconstrained physical I/O placement in the out-of-context harness.

B. Physical FPGA fixture and negative cases

The October 27 board fixture contains 3,072 bytes from the trained adapter, 256 bytes per layer: the first 64 bytes of each down_w, down_b, up_w and up_b slice. An on-FPGA checker compares every encryption beat/tag and valid decryption beat against an independent Python oracle. Each core count was programmed and captured twice through JTAG ILA, passing all ten cases in both runs with exact agreement with simulated case cycle counts. The complete 2,397,696-byte adapter was not transferred through the FPGA in this test [20].

Cores	Board MHz	Setup / hold ns	Encrypt cycles / us	Decrypt cycles / us
1	90	+0.963 / +0.038	1,021 / 11.344	1,009 / 11.211
2	90	+0.776 / +0.032	511 / 5.678	505 / 5.611
4	75	+0.306 / +0.028	307 / 4.093	304 / 4.053

Table XVI. Measured board cycle counters for the 3,072-byte fixture. Times use nominal MMCM frequencies without external calibration. Every configuration passes 10/10 cases twice [20].

The ten cases cover ideal encryption/decryption, input gaps and output stalls in each direction, corrupted final-layer tag, ciphertext tampering, incorrect chunk identity, abort, reset during traffic, and successful decryption after reset. Injected comparator faults were detected in simulation before bitstream generation. Provisional plaintext terminates at the trusted internal checker; only test status is externally visible. Fixture nonces are public repeat-test values, not production encryption nonces.

The final board harness fails 100 MHz for one and two cores and fails 90 MHz for four; it passes at 90, 90 and 75 MHz respectively. Its fixture ROM, checker and debug IP introduce different paths from the integration harness, so the two clock tables must remain separate. Saved physical attempts have zero routing errors, missing clocks, unconstrained internal endpoints, multiple-clock pins and combinational loops. The selected routes pass debug-IP bus-skew constraints. Button and visual LED paths are excluded from external I/O timing; raw warnings are retained. No flash programming, physical full-file transfer, DDR2 measurement or authenticated compute integration is established by these tests.

XVII. Limitations and next work

This version is an unprotected cryptographic core. It does not prevent power or electromagnetic side-channel analysis, and it does not detect injected faults. Kandi et al. describe threshold and redundancy-based hardware options that could be studied in subsequent work [3]. Sections XIII and XIV measure the implemented unrolling and buffering tradeoff, including folded absorption. End-to-end benefits from memory transport and lossless compression require additional system measurements. Independent lanes are evaluated in Section XVI; the pipeline control experiment is in Section XV. The external quarantine buffer is also a required part of safe decryption, not an optional application feature.

XVIII. Findings

For this adapter, eliminating controller handoffs halves the buffered three-round core's cycle count without removing rounds. The folded core requires 449,581 cycles per direction, matches the reference ciphertext and tag, and reconstructs the complete checkpoint exactly. Its three-clock full-block interval is supported by simulation; its 100 MHz closure is a routed implementation result. Physical full-file transfer, memory bandwidth and authenticated compute integration remain unmeasured. Both the AD empty-terminator defect and legacy decryption tag exposure are fixed in this branch, with focused regressions and negative controls.

The dependent p8 experiment identifies a different bottleneck: controller fixes reduce routed-period-derived block latency from 70.8 to 49.8 ns at the same 12 clocks, while round splitting adds latency. Multicore simulation achieves approximately 2.000 and 3.001 times one-core throughput for two and four cores under ideal service. The later physical tests validate only the 3,072-byte fixture, with board clocks and cycle counts reported separately from full-file simulation and modeled memory bandwidth.

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Appendix A Reproducing the environment and training

Commands are grouped by the shell in which they run. WSL installation uses administrator PowerShell; the project and training commands run in Ubuntu. The following commands document the completed setup. Package versions observed in Table II should be retained when reproducing the numerical results, because an unpinned requirements installation may select newer releases.

A1 Windows PowerShell setup

```
wsl --install -d Ubuntu-22.04
# Restart Windows if requested, then continue:
wsl --update
wsl --list --verbose
wsl -d Ubuntu-22.04
```

Create the Ubuntu username and password on first launch. Confirm VERSION 2 in the distribution list. Password characters are intentionally invisible at the prompt. WSL GPU access was then checked inside Ubuntu with nvidia-smi.

A2 Ubuntu project and Python environment

```
cd ~
sudo apt update
sudo apt install -y git build-essential python3-pip python3-venv
mkdir -p ~/projects
cd ~/projects
git clone https://github.com/SPIRE-GMU/SAGE_Accelerator.git
cd SAGE_Accelerator
git checkout 4e586d80b61ad7577c64391202a8c43104728f1d
cd llmc-gpt2
python3 -m venv .venv
source .venv/bin/activate
python -m pip install --upgrade pip
python -m pip install torch --index-url https://download.pytorch.org/whl/cu128
python -m pip install -r requirements.txt
nvidia-smi
```

The checkout command above pins the source revision used in this experiment. For later sessions, return to llmc-gpt2 and reactivate .venv. No Linux NVIDIA display driver or separate CUDA compiler installation was needed for the PyTorch workflow [8], [10].

A3 CUDA calculation check

```
python3 - <<'PY'
import torch
print(torch.__version__)
assert torch.cuda.is_available()
print(torch.cuda.get_device_name(0))
x = torch.randn(256, 256, device="cuda")
y = x @ x
torch.cuda.synchronize()
print(y.shape)
PY
```

The closing PY must be on a line by itself. During setup, visible bracketed-paste markers were resolved by typing bind 'set enable-bracketed-paste off' in Bash. This changes terminal input behavior and has no effect on model results.

Appendix A Dataset and correctness checks

```
python3 dev/data/tinyshakespeare.py
python3 train_gpt2.py --device cuda --model gpt2 \
  --num_iterations 0 --write_tensors 1 --batch_size 4 \
  --sequence_length 64 --total_batch_size 256
make test_gpt2
OMP_NUM_THREADS=8 ./test_gpt2
```

Require overall okay: 1 from the C test. Use batch size 4 for this reference because the ten expected losses in test_gpt2.c are fixed to that workload. A zero-update reference run reports nanms because it has no training durations to average.

A4 Final weight export added after the training loop

Append the following block at the end of train_gpt2.py, inside its main block. Its first line has four leading spaces. It saves updated FP32 weights and selects actual weight bytes after the 1,024-byte model header.

```
    if master_process and args.num_iterations > 0:
        export_dir = args.output_dir or "."
        os.makedirs(export_dir, exist_ok=True)
        export_path = os.path.join(export_dir, "gpt2_trained_fp32.bin")
        write_model(raw_model, export_path, dtype="float32")
        file_bytes = os.path.getsize(export_path)
        print0(f"Trained checkpoint: {file_bytes} bytes")
        print0(f"Weight payload including vocabulary padding: {file_bytes - 1024} bytes")
        with open(export_path, "rb") as source:
            source.seek(1024)
            sample = source.read(1024)
        assert len(sample) == 1024
        with open(os.path.join(export_dir, "weights_1KiB.bin"), "wb") as target:
            target.write(sample)
        print0("Exported 1024 weight bytes for Ascon")
```

```
python3 -m py_compile train_gpt2.py
```

The export is guarded by a positive iteration count so a reference-only run cannot be mislabeled as trained weights. The original before-training bridge files remain distinct from the final trained checkpoint.

Appendix A Training transfer and XSim execution

A5 Ubuntu GPU fine-tuning command

```
python3 train_gpt2.py \
  --device cuda --model gpt2 \
  --input_bin dev/data/tinyshakespeare/tiny_shakespeare_train.bin \
  --input_val_bin dev/data/tinyshakespeare/tiny_shakespeare_val.bin \
  --batch_size 1 --sequence_length 64 --total_batch_size 256 \
  --num_iterations 40 --overfit_single_batch 0 \
  --val_loss_every 10 --val_max_steps 5 --write_tensors 0 \
  --output_dir runs/ascon_demo_40
```

A6 Inspect and transfer the exported sample

```
stat -c "%n %s bytes" runs/ascon_demo_40/gpt2_trained_fp32.bin
stat -c "%n %s bytes" runs/ascon_demo_40/weights_1KiB.bin
sha256sum runs/ascon_demo_40/weights_1KiB.bin
cp -i runs/ascon_demo_40/weights_1KiB.bin \
  /mnt/c/Users/Symok/Downloads/gpt2_trained_weights_1KiB.bin
```

A7 Windows PowerShell VHDL simulation

```
cd "C:\Users\Symok\Downloads\ascon-aead128-vhdl-main"
python verification/gpt2_weights_demo.py `
  "C:\Users\Symok\Downloads\gpt2_trained_weights_1KiB.bin"
```

The runner uses Vivado on PATH or the installed C:\Xilinx\Vivado\2023.2\bin\vivado.bat. It first validates the Python oracle and then invokes XSim using a generated testbench. The project and results are isolated under build/gpt2_weights_demo. Before rerunning, archive that output directory if the previous receipts must be retained.

Artifact	Purpose
report.json and inputs.json	Inputs, sample hash, timing definition, and verified result
simulation_results.txt	VHDL ENCRYPT and DECRYPT PASS lines and 801-cycle counts
full_output.txt	All 64 output blocks in each direction
ciphertext.bin and tag.bin	1,024-byte ciphertext and 16-byte authentication tag
recovered_weights.bin	Byte-exact decryption result
vivado_console.log	Compile, elaboration, and simulation receipts
runs/ascon_demo_40/main.log	Ubuntu training and validation losses

The training output, source revision, installed software versions, sample hash, and simulation receipts jointly establish the provenance of this experiment. Reported training memory is PyTorch allocation; reported encryption time is simulation latency. Neither is a physical board measurement.